Fiscal Year:	FY 2021	**Task Last Updated:**	FY 12/04/2020
PI Name:	Jaing, Crystal Ph.D.		
Project Title:	International Space Station – Microbial Observatory of Pathogenic Virus, Bacteria, and Fungi (ISS-MOP) Project		
Division Name:	Human Research, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	94550-9698	**Congressional District:**	15
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	05/01/2015	**End Date:**	12/31/2021
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 12/31/2021 per F. Hernandez/ARC (Ed., 11/19/20) NOTE: End date changed to 12/31/2020 per PI and NSSC information (Ed., 11/26/19) NOTE: End date changed to 12/31/2019 per F. Hernandez/ARC (Ed., 4/18/19) NOTE change in grant number and end date per NSSC information and F. Hernandez/ARC; note grant 80NSSC18K0113 has period of performance of 10/20/2017-4/30/2019 (Ed., 1/18/2018) NOTE: Extended to 3/31/2019 per F. Hernandez/ARC (Ed., 8/22/17)		
Key Personnel Changes/Previous PI:	No changes.		
COI Name (Institution):	Mehta, Satish Ph.D. (KBR/NASA Johnson Space Center) Pierson, Duane Ph.D. (NASA Johnson Space Center) Smith, David Ph.D. (NASA Ames Research Center) Venkateswaran, Kasthuri Ph.D. (NASA Jet Propulsion Laboratory)		
Grant/Contract No.:	80NSSC18K0113 ; NNX15AJ29G		
Performance Goal No.:			

Performance Goal Text:

[Ed. note (January 2018): change in grant number and end date per NSSC information and NASA Ames Research Center; note grant 80NSSC18K0113 has period of performance of 10/20/2017-4/30/2019; Ed. update 12/11/2020--end date is now 12/31/2021]
International Space Station – Microbial Observatory of Pathogenic Virus, Bacteria, and Fungi (ISS-MOP) project will generate the microbial censuses associated with various ISS modules using traditional culture-based methods and state-of-the art molecular techniques. The objectives of the ISS-MOP project are to measure presence of viral and select bacterial and fungal pathogens and correlate their presence on crew. The effects of the spaceflight environment on viral and microbial pathogen dynamics are largely unknown and represent both a significant gap in our knowledge and an important opportunity to study the presence of microbial populations and predict health risks during long-term space exploration.

Task Description:

The hypothesis of the ISS-MOP project is that certain short-living viruses associated with inert environmental spaces in the closed habitat of the ISS can have an adverse influence on crew health. To test this hypothesis the team will catalog viruses associated with surfaces and air of the ISS modules using the all-encompassing Lawrence Livermore Microbial Detection Array (LLMDA), which contains probes to detect not only viruses but also clinically important human bacterial and fungal pathogens. In addition, skin and saliva samples from crewmembers will be requested to test the hypothesis.

There are three aims proposed in this ISS-MOP project. While the cultivation-based assay (Aim 1) provides a valid estimation of microbial presence, it offers only a limited assessment of the phylogenetic/pathogenic diversity and physiological breadth of the microbial population present. Therefore, the team will use the novel and comprehensive LLMDA microarray, which is specifically designed to elucidate the pathogenic viral and microbial diversity profiles of complex samples to at least the species level (Aim 2). In addition, we will characterize the virulence and antibiotic resistance mechanisms associated with these pathogens from the ISS by developing a new Virulence and Antibiotic Resistance gene array and analyze the samples on this array. The final deliverable of the proposed project to NASA will be the ISS-MOP dataset, a compilation of all of the genomic sequences and genetic information of viruses and microbes encountered on and within the ISS habitat (Aim 3).

The objectives of the ISS-MOP project directly respond to the National Research Council Committee for the Decadal Survey on Biological and Physical Sciences in Space recommendation P1 where it states, “to establish a microbial observatory program on the ISS to conduct long-term, multigenerational studies of microbial population dynamics.” The ISS-MOP project will also provide answers to NASA Space Biology Guiding Questions: How do alterations in gravity affect in microbial cells (CMM1); Does the enhanced virulence or other genetic changes observed with microbial cultures flown in space occur in other species (CMM4); and What are the genetic changes underlying physiological responses to spaceflight demonstrated in biological systems (CMM5).

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The effects of the spaceflight environment on viral and microbial pathogen dynamics are largely unknown and represent both a significant gap in our knowledge and an important opportunity to study the presence of microbial populations and predict health risks during long-term space exploration. International Space Station – Microbial Observatory of Pathogenic Virus, Bacteria, and Fungi (ISS-MOP) project will generate the microbial censuses associated with crewmembers and various ISS modules using both traditional culture-based methods and state-of-the art molecular techniques to a degree that has not previously been possible.

Task Progress:

[Ed. note--Progress as of November 2020]
The International Space Station (ISS) is a uniquely enclosed environment that has been continuously occupied for the last two decades. Throughout its operation, protecting the health of the astronauts on-board has been a high priority. The human microbiome plays a significant role in maintaining human health, and disruptions in the microbiome have been linked to various diseases. To evaluate the effects of spaceflight on the environmental and human microbiome, surface swabs were collected from five consecutive flights. Body swabs and saliva samples were collected from four ISS astronauts on consecutive expeditions. Samples were analyzed using shotgun metagenomic sequencing and microarrays to characterize the microbial biodiversity before, during, and after the astronauts’ time onboard the ISS. Samples were analyzed using metagenomic sequencing and DNA microarray and evaluated at an individual and population level to identify changes in microbial diversity and abundance. No significant changes in the number or relative abundance of taxa were observed between collection time points when samples from all four astronauts were analyzed together. When the astronauts’ saliva samples were analyzed individually, the saliva samples of some astronauts showed significant changes in the relative abundance of taxa during and after spaceflight. The relative abundance of Prevotella in saliva samples increased during two astronauts’ time onboard the ISS while the relative abundance of other commensal taxa such as Neisseria, Rothia, and Haemophilus decreased.

We have performed microbiological culture of environmental surface wipe and air samples from the five flights (F4-8). Overall, we observed bacterial and fungal growth from surface samples from all flights. The concentration ranged from 10^2 to 10^7 CFU/m2. There were no differences in the bacterial read counts between Flights 4-7, in both the propidium monoazide (PMA) and non-PMA treated samples. However, there were differences in fungal reads, with the highest fungal counts obtained from F6, in both the non-PMA and PMA treated samples.

We analyzed the microbial profiles from all eight surfaces across four different environmental flights F4-F7 using metagenomics sequencing. Most of the top 12 most abundant species such as Propionibacterium acnes, six Staphylococcus spp., Haemophilus parainfluenzae, Rothia mucilaginosa, Corynebacterium sp. GD7., and Streptococcus mitis are all part of the normal oral microbiome.

The combination of both shotgun metagenomic sequencing and microarrays showed the benefit of both technologies in monitoring microbes on-board the ISS. There were some changes in each astronaut’s microbiome during spaceflight, but these changes were not universal for all four astronauts. These results provide insight for future ISS microbial monitoring studies and targets for antimicrobial resistance screenings.

Bibliography Type:	Description: (Last Updated: 04/07/2023)
Articles in Peer-reviewed Journals	Avila-Herrera A, Thissen J, Urbaniak C, Be NA, Smith DJ, Karouia F, Mehta S, Venkateswaran K, Jaing C. "Crewmember microbiome may influence microbial composition of ISS habitable surfaces." PLoS One. 2020 Apr 29;15(4):e0231838. https://doi.org/10.1371/journal.pone.0231838 ; PMID: 32348348; PMCID: PMC7190111 , Apr-2020
Articles in Peer-reviewed Journals	Urbaniak C, Lorenzi H, Thissen J, Jaing C, Crucian B, Sams C, Pierson D, Venkateswaran K, Mehta S. "The influence of spaceflight on the astronaut salivary microbiome and the search for a microbiome biomarker for viral reactivation." Microbiome. 2020 Apr 20;8(1):56. https://doi.org/10.1186/s40168-020-00830-z ; PMID: 32312311; PMCID: PMC7171750 , Apr-2020